



Title of the Dissertation

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2026

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Abstract

Acknowledgements

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Chapter 1 Introduction

Prediction markets and option markets both attach prices to future states of the world, but they do so through different securities, participants, and market institutions. A binary prediction contract paying one dollar if Bitcoin finishes above a threshold can be read as an event probability. A strip of options with the same underlying and maturity can be used to recover state prices and, therefore, a risk-neutral probability for the same terminal region. When these objects refer to the same asset and terminal date, their comparison offers an unusually direct way to study how two segmented markets price uncertainty.

The comparison is economically useful precisely because equality is not automatic. A Polymarket price reflects the beliefs, risk preferences, collateral constraints, and liquidity of prediction-market traders. A Deribit-implied probability is recovered under a risk-neutral measure and embeds option-market risk premia, smile dynamics, and estimation choices. The contracts can also differ in their exact settlement reference and terminal time. A measured wedge therefore need not be an error by either venue, and it is not by itself an arbitrage. It is instead a cross-market measurement object whose magnitude, shape, and dynamics can reveal where the two probability systems agree and where market structure or pricing measures keep them apart.

This question has become empirically relevant as cryptocurrency prediction markets and listed options have both developed repeated, short-horizon contracts on BTC and ETH. The nearest existing study, Portnaya (2026), compares Polymarket Bitcoin threshold prices with option-implied digital values for a small set of exactly matched contracts. That evidence establishes a persistent positive pointwise wedge, especially for low option-implied probabilities, but leaves two related issues open. First, a single threshold cannot show whether the markets disagree about the centre, width, or tails of the full terminal distribution. Second, persistence in a price gap does not establish whether the two venues are integrated intraday or which venue incorporates information first. This dissertation addresses those issues by extending the comparison from isolated binary claims to repeated event distributions and by separating distributional consistency from higher-frequency integration.

1.1 Research questions

The dissertation asks three empirical questions:

1. **RQ1: Distributional consistency.** Do Polymarket and Deribit imply consistent full terminal-price distributions for matched BTC and ETH events?

2. **RQ2: Intraday integration.** Are near-the-money survival probabilities integrated across the two venues at an observable intraday frequency?
3. **RQ3: Directional price discovery.** Can the available data identify which venue incorporates information first?

The wording of RQ3 is deliberate. Price discovery is an identification problem, not a requirement that one venue must be ranked above the other. If the available frequency, liquidity, and measurement quality do not separate the directions reliably, *unidentified* is an empirical answer rather than a missing result. Similarly, tradability is not formulated as a fourth research question. An execution-aware test would require contemporaneous quotes, order-book depth, fees, funding, collateral, and exact settlement-risk mapping, none of which is present at the required granularity in the current data.

1.2 Empirical design

The empirical design has two complementary tracks. Track A compares complete terminal-price distributions at the daily frequency. The canonical universe contains 124 BTC and ETH price events, of which 79 bucket-distribution events define mutually exclusive terminal-price cells. For each eligible event-day, Polymarket cell prices are normalised to form a probability vector. Deribit option prices are then fitted with a non-negative, regularised state-probability model and mapped to the identical Polymarket bucket grid. The comparison measures cell-level L1 and L2 distance and lower-dimensional differences in location, spread, skew, and tail mass. Quality gates for Polymarket partition sums, Deribit strike freshness, option repricing, fitted forward error, and probability mass produce a final Track A sample of 294 event-days from 61 events, or 3114 cell-day observations.

Track B asks a narrower question at a higher frequency. One reference boundary K^* is chosen ex ante for each bucket event using the first clean post-warm-up Polymarket distribution. The Polymarket survival probability above that boundary is compared with a local Deribit digital estimate constructed from fresh call strikes bracketing K^* . Hourly option trading is too sparse and noisy to support directional inference, so hourly results are retained as a feasibility diagnostic. The primary integration tests use six-hour blocks, each formed from the last traded close for each instrument within the block. This frequency improves strike and parity-pair availability but does not remove within-block non-synchronicity. The method therefore tests integration at six hours and does not claim to recover sub-six-hour leadership.

This two-track structure separates two questions that would otherwise be confounded. Track A has enough cross-strike information to compare distributional shape but only at daily frequency. Track B has an intraday time dimension but only for a local survival

probability. The design does not pretend that a sparse hourly option grid supports a full intraday risk-neutral density.

1.3 Main findings

Three findings organise the dissertation. First, the two venues agree closely on the centre of the terminal distribution. In the controlled location regression, the intercept is 0.000572 with $p = 0.8015$, and the model explains almost none of the location variation. The data therefore do not support a simple claim that Polymarket is systematically more bullish or bearish than Deribit.

Second, disagreement is concentrated in distributional shape. Relative absolute divergence is approximately 0.86 in the tail cells and 0.57 in the body. Polymarket also produces a wider distribution under the baseline and light-to-moderate regularisation settings: the baseline median spread difference is 0.0040, and Polymarket is wider on about 70.7% of event-days. This width result is conditional, however. Under the heaviest smoothing setting examined, the common-sample median spread difference falls to 0.0008 and the Polymarket-wider share to 0.526. The stable conclusion is therefore centre alignment with material tail-relative disagreement; the sign and magnitude of the width wedge remain regularisation-dependent.

Third, the two markets are integrated around the local reference strike at the frequency the data can measure. At six hours, the survival-probability level correlation is 0.912, and the contemporaneous correlation of changes is 0.535 across 889 change pairs from 78 events. The lagged cross-correlations are small, while pooled regressions are asymmetric in a way that is consistent with differential measurement error in the noisier Deribit series. These diagnostics support contemporaneous six-hour integration but do not identify a leading venue. RQ3 is therefore answered negatively: directional leadership cannot be recovered reliably from the available OHLC panel.

1.4 Contributions and scope

The dissertation makes three contributions. The first is empirical breadth. It moves from a small set of pointwise threshold comparisons to a repeated BTC/ETH event panel and evaluates complete distributions on the event-defined grid. This makes it possible to distinguish centre agreement from width and tail disagreement rather than reducing every wedge to a single binary price.

The second contribution is methodological discipline. The design uses a full daily distribution only where the strike grid supports it and a local intraday digital where it does not. It also allows sensitivity checks to qualify the headline: the spread wedge is not

promoted as sign-invariant when heavy regularisation attenuates it. The same principle applies to price discovery. Lead–lag regressions are not converted into a venue ranking when symmetric diagnostics and measurement-error considerations do not support one.

The third contribution is interpretive. The combined evidence is best described as *frequency-bounded integration with a shape wedge*. The markets track the same terminal-price events and agree near the centre, yet differ more strongly in relative tail pricing. This pattern is consistent with several mechanisms, including different probability measures, favourite–longshot behaviour, participant segmentation, liquidity, horizon alignment, and settlement basis. The design does not separately identify those mechanisms, and it makes no execution-aware profit claim.

The remainder of the dissertation proceeds as follows. Chapter 2 reviews prediction-market probability aggregation, option-implied state prices, price-discovery methods, and the closest cross-market evidence. Chapter 3 defines the two-track estimators and diagnostics. Chapter 4 documents the canonical event universe, source panels, quality gates, and reference-basis audit. Chapter 5 reports the distributional and intraday findings. Chapter 6 interprets the results against the literature and considers alternative explanations. Chapter 7 states the measurement, identification, and external- validity limits before concluding.

Chapter 2 Related Work and Theoretical Background

This dissertation sits between two literatures that have, until recently, developed largely in parallel: the treatment of prediction-market prices as aggregated probability forecasts, and the extraction of risk-neutral state prices from option markets. The natural question at their intersection—whether a prediction-market probability for a threshold event agrees with the option-implied price of the corresponding digital claim—was, at the outset of this project, apparently open for cryptocurrency markets. Portnaya (2026) now provides the closest option-implied benchmark test for crypto threshold contracts, matching Polymarket Bitcoin contracts against Binance- and Deribit-implied digital values. Her work is therefore treated as the nearest antecedent to this dissertation. The argument developed below is that her design buys clean identification at the cost of breadth and leaves the direction of information flow untested. Those two gaps define the contribution of this dissertation. Accordingly, the original parity question is retained as a replication-and-extension exercise (Section 2.7), while the primary contribution is repositioned around price discovery between prediction markets and crypto options.

2.1 Prediction Markets as Probability Aggregators

The canonical justification for reading prediction-market prices as probabilities rests on the claim that trading aggregates dispersed private information into a price that approximates the market’s collective belief. Wolfers and Zitzewitz (2004) survey the early evidence and argue that, under competitive trading and binary (Arrow–Debreu) payoffs, the equilibrium price of a contract paying one unit on an event tracks the probability that the event occurs. This is the result that motivates the entire comparison undertaken here: if a Polymarket “Yes” price is a probability, it is in principle commensurable with an option-implied risk-neutral probability for the same event.

The equivalence is, however, an approximation rather than an identity, and the qualifications matter for what follows. Manski (2006) shows formally that, with heterogeneous beliefs, a prediction-market price need not equal the mean belief and may instead fall within bounds that become wide when beliefs are dispersed. A prediction-market price is therefore not a clean physical probability—a point that becomes central when it is set against an option-implied benchmark that carries its *own*, generally different, risk adjustment. The wedge between the two prices, in other words, is over-determined: it can reflect informational disagreement, but also differences in risk pricing and participant composition across the two venues.

Empirically, prediction-market prices exhibit a systematic and well-documented dis-

tortion in the tails. Snowberg and Wolfers (2010) decompose the favorite–longshot bias and conclude that it is better explained by probability misperception—traders overweighting low-probability outcomes—than by risk-loving preferences. In a contemporary, large-scale calibration study spanning Kalshi and Polymarket, Le (2026) examine roughly 292 million trades across 327,000 binary contracts and find that reading prices at face value as probabilities is systematically misleading, with predictable biases concentrated in particular regions of the probability scale. The implication for this dissertation is direct: any divergence observed against an option benchmark should be tested for concentration in the low-probability (longshot) cell, because the prediction-market literature predicts precisely such a pattern independently of any option-pricing consideration. This is the first robustness cut imposed in Chapter 5.

2.2 Option-Implied State Prices and Risk-Neutral Densities

The benchmark against which prediction-market probabilities are compared comes from the option-implied risk-neutral measure. The foundational result is due to Breeden and Litzenberger (1978): the prices of state-contingent claims are recovered from the convexity of the call-price surface in the strike. For a call price $C_t(K, \tau)$ at strike K and maturity τ , the risk-neutral density is

$$q_t(K) = e^{r\tau} \frac{\partial^2 C_t(K, \tau)}{\partial K^2},$$

and the risk-neutral probability of finishing above a threshold—the object that a Polymarket threshold contract is meant to price—is the corresponding digital value,

$$\Pr^Q(S_T > K) = -e^{r\tau} \frac{\partial C_t(K, \tau)}{\partial K}.$$

Under the Black–Scholes special case this collapses to the cash-or-nothing call,

$$P_t^{\text{fair}} = e^{-r\tau} \Phi(d_2), \quad d_2 = \frac{\ln(S_t/K) + \left(r - \frac{1}{2}\sigma^2\right)\tau}{\sigma\sqrt{\tau}},$$

which is the exact benchmark Portnaya (2026) adopts after backing σ out of Binance option mid-quotes. The gap she studies is then

$$D_t = p_t^{\text{poly}} - P_t^{\text{fair}}.$$

Two caveats from the risk-neutral-density literature bound the interpretation of any such gap. First, the recovered density is risk-neutral, not physical: option-implied mea-

sures embed a pricing kernel, so P^{fair} differs from the true probability by a risk premium that is itself state-dependent (Jackwerth, 2004). Comparing it to a prediction-market price therefore compares two market-implied objects, not a probability against ground truth. Second, the Black–Scholes digital is sensitive to the volatility input near the threshold, so the choice of implied-volatility surface (single quote versus smile-interpolated) is a methodological decision with material consequences for the measured gap—a point exploited in the robustness design of this dissertation, since the model-free Breeden and Litzenberger (1978) digital and the Black–Scholes digital need not coincide when the smile is pronounced.

2.3 The Closest Comparison: Portnaya (2026)

Portnaya (2026) is the most direct antecedent of this work and is addressed head-on rather than buried in related work. Her question is the pointwise parity question: for a Bitcoin threshold event “BTC above K at expiry,” does the Polymarket “Yes” price equal the option-implied digital value P^{fair} ? Her design is deliberately narrow in order to make the mapping exact—same strike, same maturity, hourly alignment—and her tests are appropriate for the question: a mean-gap test with HAC standard errors and a block bootstrap, an AR(1) persistence estimate, a frictions band, and a delta-hedged arbitrage proxy.

Her results are reported here as established findings to be replicated, not re-derived.¹ On the main Binance-compatible BTC contract she finds a mean gap of 5.58 percentage points ($n = 214$, $t = 6.46$, HAC confidence interval $[0.0228, 0.0889]$). Pooling three Binance-compatible BTC contracts raises the sample to $n = 287$ with a mean gap of 6.33 percentage points; a Deribit extension yields a substantially larger pooled gap of 11.05 percentage points ($n = 2585$). The divergence is persistent rather than transient, with an estimated AR(1) half-life of roughly four hours, and is largest where the option-implied probability is low and time-to-expiry is long—consistent with the longshot pattern noted in Section 2.1. Evidence for Ethereum is mixed. Her delta-hedged arbitrage proxy produces a positive pooled net PnL after costs but only marginal statistical significance ($p = 0.053$).

Her interpretation is that Polymarket prices sit systematically above the option-implied risk-neutral benchmark, particularly for low-probability threshold events, and she attributes this to a combination of retail and speculative demand inflating longshot “Yes” prices, slow arbitrage between segmented venues, and differing risk adjustments embedded on the two sides. Crucially, she does not build a structural model and does not estimate the direction of information flow.

¹All figures in this paragraph are taken from your fact-check of the source and its tables; they are reproduced for context and should be cited to Portnaya (2026), not to this dissertation.

The boundaries of her design are the entry points for this dissertation. The exact-matching requirement restricts her main Binance panel to three BTC contracts and 287 hourly observations. That is sufficient to demonstrate that a wedge *exists*, but it cannot establish whether the wedge is a general feature across events, maturities, moneyness and event types; whether it is confined to the tail; or—most importantly—which venue moves first. Her AR(1) half-life describes how fast the gap closes but not which side does the closing. This dissertation trades a degree of mapping purity for external validity and price-discovery power, addressing exactly these questions.

2.4 Price Discovery: Methods and Applications

The primary contribution of this dissertation is methodological in the price-discovery sense, so the relevant toolkit is reviewed here. When two venues trade closely related claims on the same underlying state, the empirical question is whether their prices share a stable common component and, conditional on that diagnostic, which venue impounds new information first. If the two series are non-stationary and cointegrated, the natural representation is the vector error-correction model (VECM):

$$\Delta p_t = \alpha \beta' p_{t-1} + \sum_{i=1}^k \Gamma_i \Delta p_{t-i} + \varepsilon_t, \quad \beta = (1, -1)',$$

where the cointegrating vector is shown in the frictionless one-for-one case and can be augmented with an intercept to allow a non-zero equilibrium wedge. The adjustment vector α governs which series moves when the spread deviates from that long-run relation. Two canonical decompositions sit on top of this representation. The Hasbrouck (1995) information share attributes the variance of the efficient-price innovation across venues and is bounded only when the innovations are correlated, requiring an ordering. The Gonzalo and Granger (1995) permanent–transitory (component-share) decomposition uses the orthogonal complement of α and yields a single value, but measures a different quantity—the venue’s contribution to the common factor rather than to its innovation variance.

A well-known limitation of both metrics is that they are contaminated by the relative noise of the two venues, so a venue can appear to “lead” simply because it is less noisy. Putniņš (2013) dissects exactly what each metric measures and clarifies the conditions under which they diverge. Patel et al. (2020) apply this critique in the stock-versus-options setting most analogous to the present one—options are typically the noisier, lower-volume venue, which biases naive information shares against them—and re-estimate price-discovery contributions using the information-leadership share (ILS) and information-leadership indicator (ILI), which are designed to separate leadership from relative noise.

This is directly relevant: a prediction market and a crypto option market differ sharply in liquidity and quoting frequency, so this dissertation treats ILS/ILI as the preferred cointegration-based specification when the diagnostics justify that framework.

The diagnostic caveat is important. Polymarket prices and option-implied digital values are bounded between zero and one, and their gap may be stationary even when the individual series are not well described as integrated processes. The empirical design therefore does not assume a VECM mechanically. It first tests stationarity and cointegration; if the VECM assumptions fail, price discovery is estimated through complementary specifications such as lead–lag correlations, Granger-style predictive regressions in changes, and local projections around large option- or prediction-market moves. This diagnostic-driven approach prevents the price-discovery result from resting on an inappropriate time-series representation.

Within prediction markets, Ng et al. (2026) provide the closest price-discovery precedent, studying common election contracts traded across Polymarket, Kalshi, PredictIt and Robinhood and finding that Polymarket leads Kalshi when liquidity is high. That result establishes that prediction markets can be the leading venue in a multi-market system, but the comparison is prediction-market-versus-prediction-market; it does not address whether a prediction market leads or lags an *option* market, which is the question this dissertation examines.

2.5 Microstructure and Data-Integrity Constraints

A price-discovery study is only as credible as the trade and quote data feeding it, and the decentralised microstructure of Polymarket imposes constraints that a naive WebSocket-based pipeline would silently violate. Dubach (2026) reconstructs Polymarket tick and order-book data at scale (on the order of 30 billion events across roughly 600 markets) and reports that trade direction inferred from the public order-book feed agrees with the on-chain ground truth only about 59% of the time. For lead–lag and order-flow analysis this makes inferred signed order flow unreliable. The methodological consequence, adopted in Chapter 4, is that trade direction should be taken from the on-chain `OrderFilled` events where signed flow is required, rather than inferred from the public feed.

A second data-integrity caveat concerns settlement. Shen et al. (2026) document off-chain matching followed by on-chain settlement reverts—“ghost” fills that appear to execute but do not settle—which affect both the measurement of executed volume and the interpretation of any apparent arbitrage. This does not threaten the price-discovery analysis directly but bounds the executability claims in the tradability analysis, and is carried into the limitations discussion accordingly.

2.6 Limits to Arbitrage and Semantic Non-Fungibility

The final strand concerns whether a measured statistical divergence is economically exploitable, and the literature gives strong reasons to expect that it is not cleanly so. Cheng et al. (2026) analyse Polymarket NBA markets using roughly 75 million order-book snapshots across 173 games and find that executable arbitrage does exist but is sharply constrained by depth, operating at retail rather than institutional scale. Gebele and Matthes (2026) examine ten venues over 2018–2025 and more than 100,000 events, finding that semantically equivalent markets continue to display execution-aware deviations of roughly two to four percent that do not close. Together these results require a strict distinction between a statistical wedge and an executable arbitrage. The present dissertation does not contain the order-book depth, fees, collateral and settlement-risk data needed for a trading test, so tradability is treated as an economic interpretation boundary rather than as an answered empirical research question.

The frictions that can sustain the wedge are specific and enumerable. They include horizon mismatch between a fixed-expiry prediction contract and an option hedge; pay-off mismatch between a binary prediction claim and a continuous option position used to replicate it; oracle and settlement risk on resolution (Shen et al., 2026); depth and liquidity limits (Cheng et al., 2026); and fees and execution timing. These mechanisms bound the interpretation in Chapters 6 and 7: even a persistent statistical wedge cannot be labelled an arbitrage without execution-aware evidence.

2.7 Research Gap and Positioning

The preceding sections locate a precise gap. Portnaya (2026) has established that a pointwise binary wedge between Polymarket and crypto-option-implied digital values exists for a narrow set of BTC threshold contracts, and that it is persistent and concentrated in the longshot tail. What remains unestablished is whether the comparison generalises beyond a handful of exactly matched contracts to a broader event panel and to distributional, rather than single-threshold, consistency; whether comparable local probabilities are integrated across the two venues at an observable intraday frequency; and whether the available data can identify which venue incorporates information first.

This dissertation is positioned against that gap as follows. *Portnaya tests whether Polymarket binary threshold prices match option-implied binary values for a small set of BTC and ETH contracts. This dissertation extends the question from pointwise binary parity to distributional consistency across a broader event panel and then tests whether a local survival probability is integrated across the two venues at an observable intraday frequency.* Concretely, the original parity question becomes a distributional replication-and-extension exercise; the price-discovery contribution is diagnostic-driven, using lead–

lag tests only at the frequency supported by the data; and the empirical design permits directional leadership to remain unidentified rather than forcing an information-share estimate on an unsuitable panel. Table 2.1 summarises the final three research questions.

Table 2.1: Positioning of the research questions relative to existing work.

RQ	Question	Closest prior work	Status
RQ1	Do prediction-market probabilities agree with option-implied state prices across a broader event panel (distributional, not pointwise)?	Portnaya (2026) (pointwise binary, narrow BTC panel)	Replication + extension
RQ2	Are near-the-money survival probabilities integrated across the two venues at an observable intraday frequency?	Portnaya (2026) (persistence); Ng et al. (2026) (PM-vs-PM integration)	Cross-market extension
RQ3	Can the available data identify which venue incorporates information first?	Patel et al. (2020); Putniņš (2013) (measurement and leadership methods)	Primary identification test

Chapter 3 Methodology

This chapter defines the empirical objects compared in the dissertation. The central difficulty is that the two venues do not quote identical securities. Polymarket quotes binary event claims over terminal BTC/ETH price regions, while Deribit quotes options whose prices imply risk-neutral state prices over the same underlying. The methodology therefore does not test a literal law of one price. It constructs comparable probability objects and then asks where the constructed objects agree, where they diverge, and whether their changes are integrated at intraday frequency.

The empirical design has two tracks. Track A is a daily full-distribution comparison. It reconstructs a complete Polymarket terminal-price distribution on each event's bucket grid and compares it with a Deribit-implied distribution obtained from a smoothed, shape-disciplined option curve. Track B is an intraday local comparison. It fixes one reference strike K^* per event and compares the survival probability $P(S_T > K^*)$ across venues. Track B is not a full intraday risk-neutral-density exercise; it is deliberately local because hourly option coverage is too sparse to support a full surface.

3.1 Notation and comparison objects

Let e index a Polymarket event, t a date or intraday bar, and S_T the underlying BTC or ETH price at the event's terminal resolution time T . For a bucket-distribution event, Polymarket defines a finite partition of the price axis,

$$\mathcal{B}_e = \{B_{e1}, \dots, B_{eJ_e}\},$$

where the first and last cells are open-ended tails and the interior cells are finite price intervals. The Polymarket probability vector on this grid is denoted

$$\mathbf{p}_{e,t}^{PM} = (p_{e,t,1}^{PM}, \dots, p_{e,t,J_e}^{PM}).$$

The Deribit-implied probability vector mapped to the same grid is

$$\mathbf{p}_{e,t}^D = (p_{e,t,1}^D, \dots, p_{e,t,J_e}^D).$$

The object of Track A is the difference between these two vectors. The object of Track B is scalar:

$$q_{e,t}^{PM} = P_{e,t}^{PM}(S_T > K_e^*), \quad q_{e,t}^D = P_{e,t}^D(S_T > K_e^*).$$

The two tracks are complementary. Track A measures distributional disagreement but only at daily frequency. Track B measures higher-frequency integration but only around one local threshold.

3.2 Polymarket probability reconstruction

3.2.1 Cell probabilities and normalisation

Each Polymarket event is represented as a collection of YES-token markets. For bucket-distribution events, the event cells form a complete partition of the terminal price axis. Let $m_{e,t,j}$ be the observed midpoint-like price for cell j at time t . The raw event probability mass is

$$M_{e,t} = \sum_{j=1}^{J_e} m_{e,t,j}.$$

When the event passes the sum-quality gate described in Chapter 4, the normalised Polymarket probability assigned to cell j is

$$p_{e,t,j}^{PM} = \frac{m_{e,t,j}}{M_{e,t}}.$$

The raw values are retained for diagnostics, but the normalised values are used for the main Track A distribution comparison so that the vector integrates to one on the event grid. Event-days with poor partition-sum quality are removed from the main comparison sample rather than corrected silently.

For point-threshold events, the Polymarket price is not a full distribution. It is a direct YES price for a statement of the form $S_T > K$. These events can therefore support a local survival probability in Track B, but they cannot support Track A full-distribution comparisons.

3.2.2 Track B survival from Polymarket cells

For bucket-distribution events, the Track B reference strike K_e^* is a finite Polymarket boundary chosen ex ante. The selection rule uses the first clean post-warm-up Polymarket snapshot and chooses the boundary whose survival probability is closest to one-half:

$$K_e^* = \arg \min_{K \in \mathcal{K}_e} \left| \sum_{j: B_{ej} \subseteq (K, \infty)} p_{e,t_0,j}^{PM} - \frac{1}{2} \right|,$$

where $\mathcal{K}_e$ is the set of finite event boundaries and t_0 is the first clean post-warm-up snapshot. After K_e^* is fixed, the Polymarket survival probability is

$$q_{e,t}^{PM} = \sum_{j: B_{ej} \subseteq (K_e^*, \infty)} p_{e,t,j}^{PM}.$$

For point-threshold events, K_e^* is the market-defined threshold and $q_{e,t}^{PM}$ is the YES price itself. These events are not part of the primary Track B sample because their survival series are often saturated near zero or one, leaving too little time-series variation for lead-lag analysis. They are kept only as an explicitly separated extension.

3.3 Deribit-implied probability reconstruction

3.3.1 Coin-denominated option prices and parity-implied spot

Deribit option prices are quoted in the underlying coin. To compare them with Polymarket USD price regions, the pipeline infers a same-period spot level from put-call parity. For a call and put with the same strike K and same expiry, the working parity relation in coin units is

$$C^{coin}(K) - P^{coin}(K) \approx 1 - \frac{K}{S}.$$

Solving for S gives the strike-level implied spot

$$\hat{S}(K) = \frac{K}{1 - (C^{coin}(K) - P^{coin}(K))}.$$

The fitted spot used for an event-day or intraday bar is the median of valid same-period parity-implied spots after basic denominator and positivity filters. This proxy is necessary because the raw OHLC files do not retain historical Deribit index levels. The working relation abstracts from interest-rate, funding, and discounting terms over these short horizons; it is therefore a measurement approximation rather than an exact parity identity and is recorded as a limitation.

3.3.2 Track A state-price fit

Track A does not apply raw second differences directly to discrete option prices. Instead, for each event-day it fits a non-negative state-probability vector on a grid that combines Deribit strikes, Polymarket cell boundaries, and the parity-implied spot. Let $x_1, \dots, x_K$ denote the midpoints of the state intervals and π_k the state probability at x_k .

For each traded option i , define its payoff vector a_i in coin units. For a call with strike

K_i ,

$$a_{ik} = \frac{\max(x_k - K_i, 0)}{\hat{S}},$$

and for a put,

$$a_{ik} = \frac{\max(K_i - x_k, 0)}{\hat{S}}.$$

The fitted probabilities solve a constrained least-squares problem of the form

$$\min_{\pi \geq 0} \sum_i \left(a_i^\top \pi - y_i \right)^2 + w_\Sigma^2 \left(\mathbf{1}^\top \pi - 1 \right)^2 + w_\mu^2 \left((x/\hat{S})^\top \pi - 1 \right)^2 + w_s^2 \sum_k (\pi_k - 2\pi_{k+1} + \pi_{k+2})^2,$$

where y_i is the observed coin option price. The sum and forward penalties anchor the distribution to unit mass and to the parity-implied forward level; the second-difference penalty regularises the state probabilities. The squared weights appear because the implementation appends weighted rows to a least-squares system; reported smoothness settings refer to the row weight w_s , not to w_s^2 . After fitting, the probability vector is normalised to sum to one.

This estimator is intentionally conservative. It enforces non-negativity and avoids raw finite-difference densities, but it also makes the recovered distribution dependent on the smoothness penalty. The dissertation therefore reports spread results as smoothing-conditional and treats cell-level L1/L2 distances as secondary diagnostics rather than hard headline magnitudes.

3.3.3 Mapping the Deribit distribution to Polymarket cells

Once a state-probability vector is fitted, Deribit probability mass is mapped onto the Polymarket event grid. For cell $B_{ej} = [L_j, U_j)$,

$$p_{e,t,j}^D = \sum_{k: x_k \in [L_j, U_j)} \pi_k,$$

with the natural extension for open-ended left and right tail cells. This produces a Deribit-implied probability vector with the same dimension and cell definitions as the Polymarket vector.

Each fitted curve must pass sanity checks before entering the main Track A comparison. The checks include probability mass close to one, non-negative state probabilities, fitted-forward error within tolerance, option-repricing RMSE within tolerance, and Deribit cell probabilities summing to one after mapping. Event-days that fail these checks are retained in diagnostics but removed from the main comparison sample.

3.4 Track A distribution-divergence metrics

3.4.1 Cell-level distances

For each event-day, the primary cell-level difference is

$$d_{e,t,j} = p_{e,t,j}^{PM} - p_{e,t,j}^D.$$

The aggregate distances are

$$L1_{e,t} = \sum_j |d_{e,t,j}|, \quad L2_{e,t} = \left(\sum_j d_{e,t,j}^2 \right)^{1/2}.$$

These distances describe shape disagreement on the event bucket grid. They are not used as the main economic magnitude because they are sensitive to the smoothness of the fitted Deribit density, especially in the tails.

3.4.2 Moment and tail diagnostics

The moment diagnostics are computed on cell moneyness rather than on raw dollar price.

Let $z_{e,t,j}$ be the representative moneyness of cell j ,

$$z_{e,t,j} = \frac{\text{cell midpoint}_{e,j}}{\hat{S}_{e,t}} - 1.$$

Open-ended tail cells use a midpoint convention based on the median finite bucket width; alternative tail-midpoint conventions are reported as robustness checks. For any probability vector p , define

$$\mu(p) = \sum_j p_j z_j, \quad \sigma(p) = \left(\sum_j p_j (z_j - \mu(p))^2 \right)^{1/2},$$

and, when $\sigma(p) > 0$,

$$\gamma(p) = \sum_j p_j \left(\frac{z_j - \mu(p)}{\sigma(p)} \right)^3.$$

The dissertation refers to these as location, spread, and skew. The reported moment wedges are

$$\Delta_{e,t}^{loc} = \mu(\mathbf{p}_{e,t}^{PM}) - \mu(\mathbf{p}_{e,t}^D), \quad \Delta_{e,t}^{spr} = \sigma(\mathbf{p}_{e,t}^{PM}) - \sigma(\mathbf{p}_{e,t}^D),$$

with the analogous definition for skew.

Tail disagreement is measured separately because absolute tail probabilities are me-

chanically small. For tail cells $\mathcal{T}_e$, the raw tail divergence is

$$\sum_{j \in \mathcal{T}_e} d_{e,t,j},$$

while relative absolute divergence scales the absolute difference by the Polymarket probability in the cell,

$$r_{e,t,j} = \frac{|d_{e,t,j}|}{p_{e,t,j}^{PM}}, \quad p_{e,t,j}^{PM} > 0.$$

The log-odds divergence uses clipped probabilities to avoid infinite values:

$$\ell_{e,t,j} = \log \left(\frac{\tilde{p}_{e,t,j}^{PM}}{1 - \tilde{p}_{e,t,j}^{PM}} \right) - \log \left(\frac{\tilde{p}_{e,t,j}^D}{1 - \tilde{p}_{e,t,j}^D} \right),$$

where $\tilde{p}$ is clipped inside $(0, 1)$. In the results chapter, tail relative and log-odds metrics are preferred to raw absolute tail differences.

3.4.3 Track A regression specification

The Track A regressions are descriptive controls for observed composition, not causal estimates. The main spread specification is

$$\Delta_{e,t}^{spr} = \alpha + \beta_g G_e + \beta_\tau \tau_{e,t} + \beta_a A_e + \varepsilon_{e,t},$$

where G_e is a categorical signed horizon-gap bin with -8 hours as the reference category, $\tau_{e,t}$ is Deribit time-to-expiry in days, and A_e is an asset indicator. Standard errors are clustered by event. The same control logic is used for location and skew diagnostics.

The signed horizon gap is not interpreted causally. It is a deterministic feature of how the Polymarket resolution time aligns with the nearest Deribit expiry. It may proxy for calendar matching, expiry type, or market composition. For that reason, a narrow -8 hour sub-sample is not promoted as the headline sample; the pooled controlled specification is the primary design.

3.5 Track B local survival and integration tests

3.5.1 Deribit local call-spread digital

Track B estimates the Deribit survival probability locally around K_e^* . Within an hourly or multi-hour bar, the estimator keeps fresh traded option rows, infers a same-bar parity spot, and selects the nearest traded call strike below K_e^* and the nearest traded call strike above K_e^* . Let those strikes be $K_L < K_e^* < K_H$, with USD call values $C(K_L)$ and $C(K_H)$.

The local call-spread digital is

$$q_{e,t}^D = -\frac{C(K_H) - C(K_L)}{K_H - K_L},$$

clipped to $[0, 1]$ for reporting. Bars without a same-bar parity spot or without fresh call strikes bracketing K_e^* fail explicitly and do not enter the informative sample.

This estimator is local by design. A full hourly risk-neutral density would require many more fresh strikes across the smile and would amplify the non-synchronicity of last-trade OHLC data. The local estimator asks a narrower question: whether the two venues agree and co-move around one economically relevant threshold.

3.5.2 Informative bars and frequency choice

A Polymarket bar is informative when it passes the quality gate, has a real update, and has survival probability inside $(0.05, 0.95)$. A Deribit bar is informative when the local call-spread digital passes, has a real trade, and its survival estimate lies inside the same interval. The joint Track B panel uses bars that are informative on both sides.

Hourly Deribit local-survival changes are too noisy for directional price-discovery inference. The dissertation therefore treats hourly data as a feasibility diagnostic and uses six-hour bars as the primary measurable integration frequency. Six-hour bars are constructed from the raw hourly OHLC by keeping fresh traded rows within the block and using each instrument's last traded close inside the block. Prices are not averaged across the six hours. This improves the availability of bracketing strikes and parity pairs, but it can combine instrument closes observed at different hours within the block and therefore does not remove non-synchronicity. Sub-six-hour leadership cannot be identified from this construction.

3.5.3 Cross-correlation and pooled lead-lag regressions

For the six-hour informative panel, define changes

$$\Delta q_{e,t}^{PM} = q_{e,t}^{PM} - q_{e,t-1}^{PM}, \quad \Delta q_{e,t}^D = q_{e,t}^D - q_{e,t-1}^D.$$

The first diagnostic is the symmetric cross-correlation between Δq^{PM} and lagged or led Δq^D . This diagnostic is used because it treats the two series symmetrically and is less vulnerable to the regression asymmetry induced by measurement error.

The second diagnostic is a pair of pooled lead-lag regressions with event fixed effects:

$$\Delta q_{e,t}^{PM} = \alpha_e + \rho_{PM} \Delta q_{e,t-1}^{PM} + \beta_D \Delta q_{e,t-1}^D + u_{e,t},$$

and

$$\Delta q_{e,t}^D = \delta_e + \rho_D \Delta q_{e,t-1}^D + \beta_{PM} \Delta q_{e,t-1}^{PM} + v_{e,t}.$$

Standard errors are clustered by event. These regressions are interpreted only after checking the symmetric cross-correlations and the relative noise of the two change series. In the final P1 design, the regression asymmetry is not interpreted as venue leadership because Deribit changes are noisier and can produce errors-in-variables attenuation when placed on the right-hand side.

The methodology therefore reports Track B as a frequency-bounded integration test: level alignment and contemporaneous six-hour co-movement are identifiable; directional leadership is not.

3.6 Reproducibility and provenance

Every generated panel is written as a processed parquet or CSV file and is accompanied by a metadata record whenever the pipeline creates one. The paper-facing tables are also indexed in a table-source metadata file that maps each table to its generating script, input panel, output table, sample gate, and intended paper use. The frozen paper artifacts were generated before the project was placed under Git, so their metadata records `git_commit` as unavailable rather than fabricating a hash. The folder is now a repository, but regenerating provenance is an engineering step outside the present empirical freeze; until that rerun, the script and input-file mappings are the operative provenance records.

The key methodological discipline is that results are not promoted beyond what their diagnostics support. The spread wedge is reported as conditional on RND smoothness; the horizon-gap gradient is reported as composition, not causality; point-threshold markets are excluded from primary Track B because of saturation; and Track B leadership is reported as unidentified because the data support integration but not a robust direction of information flow.

Chapter 4 Data

This chapter documents the construction of the empirical panels used in the two tracks of the study. Track A compares full terminal-price distributions implied by Polymarket against risk-neutral densities recovered from Deribit options at the daily frequency. Track B compares a single at-the-money survival probability across the two venues at intraday frequency. Both tracks are built from a single *canonical event universe* so that sample definitions are fixed once and reused, rather than being re-selected at each stage. Section 4.1 describes the raw sources; Sections 4.2–4.4 describe the venue-level panels; Sections 4.5 and 4.6 describe the two comparison samples and their quality gates; and Section 4.7 states the selection and measurement caveats that constrain interpretation in later chapters.

All row counts reported below are the frozen outputs of the data pipeline and are reproduced from the pipeline provenance records. Where a gate removes observations, both the surviving count and the removed count are reported so that attrition is transparent.

4.1 Data sources

The study uses two primary sources, both accessed over public interfaces. No proprietary order-book or trade-level feed was available, and this constraint shapes several of the methodological choices discussed in Chapter 3 and the limitations in Chapter 7.

Table 4.1: Primary data sources.

Venue	Interface	Content used
Polymarket	Gamma API (<code>markets</code> , <code>prices-history</code>)	Event and market metadata; sampled per-outcome price histories for all resolved BTC/ETH monthly price events in the sample window.
Deribit	REST API (<code>get_tradingview_chart_data</code>)	Historical option OHLC at daily (daily) resolution for Track A and at hourly (60) resolution for Track B, per option instrument.

Polymarket outcome prices are interpreted as market-implied probabilities. Deribit option prices are coin-denominated (quoted in BTC or ETH); the conversion to a USD reference and the recovery of state probabilities from these prices are described in Chapter 3. Because historical Deribit index levels are not retained in the raw OHLC files, the same-day spot used for the coin-to-USD conversion is inferred from same-period call–put

parity rather than taken from the exchange index; the consequences of this are recorded in Section 4.7 and Chapter 7.

4.2 Canonical event universe

The sampling unit is a Polymarket *event*: a set of mutually exclusive outcome markets referencing the terminal price of BTC or ETH at a fixed resolution date. The canonical universe contains 124 target events (60 BTC, 64 ETH). Each event is decomposed into *cells*, where a cell is a single price region with an associated YES token. The universe contains 1338 cells, partitioned by type as 687 interior buckets, 493 point-threshold (“above X ”) cells, and 79 left-tail and 79 right-tail cells. Every cell has a non-missing YES token identifier (0 missing), which is a precondition for reconstructing probabilities from price histories.

Two overlapping eligibility flags are defined on this universe and drive the two tracks:

- **Track A (bucket-distribution) eligibility:** 79 events whose interior buckets and tail cells form a complete partition of the price axis, enabling a full distribution to be reconstructed on the event’s own bucket grid.
- **Track B eligibility:** all 124 events, comprising the 79 bucket-distribution events and 45 point-threshold events, each of which admits a single reference strike K^* (Section 4.6).

Fixing this universe once means that every downstream Polymarket, Deribit, Track A, and Track B panel refers to the same events and the same cell definitions, so that later attrition is attributable entirely to data-quality gates rather than to changes in the sampling frame.

4.3 Polymarket probability panel

Price histories for all 124 events were downloaded and reconstructed into distribution panels. The raw histories comprise 191 479 rows. Because the individual cell time stamps within an event differ by a few seconds, processed observations are aligned to the UTC hour while the raw observed time stamp is retained. This yields an hourly panel of 122 923 rows and a daily panel of 5677 rows across the 79 Track A events.

Not all events maintain an outcome-probability sum close to one at all times, so an event-level quality diagnostic is applied: after a warm-up period, an event is a main candidate if a sufficiently high share of complete hourly bars has an outcome-probability sum in $[0.9, 1.1]$ and daily coverage is adequate. Of the 79 Track A events, 67 pass as main candidates, 8 are borderline on the sum criterion, and 4 fail. Only the 67 pass-main

events enter the Track A main comparison; borderline and failing events are reserved for appendix discussion.

4.4 Deribit option panel

4.4.1 Daily panel (Track A)

For the 79 Track A events, an option strike grid was constructed and daily OHLC was downloaded per instrument, producing 25 315 option-day rows and 675 event-day quality rows covering all 79 events. The median number of distinct traded strikes per event-day is 17. The distribution of the strike count is effectively bimodal: an event-day has either zero traded strikes or at least ten. The 79 zero-strike event-days correspond to exactly one day per event and are the Deribit expiry dates, on which the daily bar is unsuitable for curve reconstruction; these are removed by the coverage gate and do not enter the comparison sample.

Because a daily OHLC bar records only whether an instrument traded on a given day and not the intraday time of trade, cross-strike synchronicity cannot be measured from time stamps at this resolution. Non-synchronicity is instead captured by the share of stale bars in the daily grid, defined as the fraction of grid strikes whose price was carried forward without a fresh update; this staleness measure is the operative quality gate for Track A (Section 4.5).

4.4.2 Hourly panel (Track B)

For the 79 bucket-distribution events, hourly (60) option OHLC was downloaded, producing 491 531 rows and 13 119 hourly quality bars. This panel supports the local survival-probability estimator used in Track B and is written to a separate quality file so that it does not overwrite the daily Track A panel.

4.5 Track A comparison sample and quality gates

The Track A comparison sample is obtained by joining the Polymarket daily distribution snapshots to the Deribit daily grid on event and UTC date and then applying a sequence of gates. Table 4.2 reports the funnel.

The final Track A comparison sample contains 294 event-days from 61 events, corresponding to 3114 cell-day rows. Two gates warrant comment. The staleness gate (stale share ≤ 0.30) removes 105 of the 406 event-days; because it conditions on the freshness of the Deribit grid, it is not neutral with respect to liquidity (Section 4.7). The curve quality gate removes a further 7 event-days on which the fitted state prices fail to

Table 4.2: Track A sample funnel. Each row applies the stated gate to the previous row; both event-days and distinct events are reported.

Stage	Event-days	Events
Polymarket–Deribit joinable event-days	531	79
Legacy PM + Deribit min-8 strike gate	406	67
Curve-input gate: stale share ≤ 0.30 and ≥ 8 fresh strikes	301	61
Curve fit success	301	61
Curve quality pass (option repricing within tolerance)	294	61

reprice the input options within tolerance; the non-negativity and unit-mass properties of the fitted densities are enforced by construction and are not the reason for these failures.

Because the horizon of a Polymarket event and the nearest Deribit expiry do not coincide exactly, each event carries a signed horizon gap between its resolution time and the matched Deribit expiry. The absolute gap ranges from 8 to 40 hours with a median of 32 hours; it takes four discrete signed values (-32 , -8 , $+16$, $+40$ hours) that map one-to-one onto calendar-day offsets. This gap is a structural feature of the venue calendars rather than a data defect, and it is carried as a control in Chapter 5 rather than eliminated.

4.6 Track B survival panels

Track B compares the survival probability $P(S_T > K^*)$ across venues at a single reference strike K^* . For bucket-distribution events, K^* is chosen ex ante to be close to at-the-money at the start of the event. For point-threshold events, K^* is the market-defined threshold itself. A reference strike is available for all 124 events (79 bucket-distribution, 45 point-threshold). For bucket-distribution events K^* can be placed near the centre of the outcome grid, and the initial survival probability has median 0.490 with a median distance to one-half of 0.044; for point-threshold events the initial survival probability is far more dispersed (median distance to one-half of 0.265).

4.6.1 Polymarket survival panel

The Polymarket hourly survival panel contains 18 225 rows across all 124 events, of which 16 700 pass the partition and warm-up quality gates. An hour is treated as *informative* only if it passes quality, carries a real update, and has a survival probability strictly inside $(0.05, 0.95)$; saturated hours near zero or one carry no time-series information for a lead-lag comparison. The two market types differ sharply on this measure: bucket-distribution events have a saturated share of 0.090, while point-threshold events have a saturated share of 0.421. On a per-event basis, 73 of 79 bucket-distribution events retain at least 72 informative hours, whereas only 25 of 45 point-threshold events do, and 4 have

none. Point-threshold events are therefore excluded from the primary Track B sample and retained only as an external-validity extension; the primary Track B sample is the 79 bucket-distribution events.

4.6.2 Deribit local survival panel

The Deribit survival probability at K^* is estimated locally, from a call-spread digital using the two traded call strikes that bracket K^* within an OHLC bar, rather than from a full option-implied density. At hourly resolution this estimator is severely coverage-constrained: of the 13 119 hourly bars, only 4913 pass and 4168 are informative, the binding failures being the absence of fresh call strikes bracketing K^* and the absence of a same-hour call-put pair for spot inference. After joining to the Polymarket informative hours, only 2973 joint-informative hours remain. Only 2 events retain at least 72 total joint-informative hours, and no event has a contiguous joint-informative run of 24 hours or more; the median longest contiguous run is only 6 hours.

Re-estimating the Deribit digital on six-hour bars aggregated from the same raw hourly OHLC materially improves coverage: 1821 of 2226 six-hour bars pass and 1569 are informative. The joined six-hour panel yields 1121 joint-informative rows across the 79-event primary sample; the contemporaneous change diagnostic uses 889 usable consecutive change pairs from 78 events. The choice of the six-hour frequency, and its consequences for what can and cannot be identified, are established in Chapter 5; here it suffices to note that the intraday comparison is feasible only at this coarser frequency.

4.7 Selection and measurement caveats

Three properties of the data constrain the interpretation of later results and are stated here so that the empirical chapters can refer back to them.

Selection on treatment from the staleness gate. The Track A staleness gate removes low-freshness Deribit grids, which are systematically the less-liquid events. The 6 events removed entirely by the gate have a median traded volume of 145 844, against 531 367 for the events that survive, a factor of roughly 3.6. The final comparison sample is therefore selected toward more liquid events and is not population-representative. The direction of the resulting bias in the probability wedge is not identified: less-liquid events may exhibit larger segmentation effects, but they may also contain more measurement error, and the excluded outcomes are not observed cleanly enough to separate those channels.

Horizon and reference-basis mismatch. The Polymarket resolution instant and the matched Deribit expiry differ by 8–40 hours, and the two venues settle against different references: Polymarket resolves on a centralised-exchange spot print, while Deribit settles against its own index. Aligning the comparison in time does not remove the reference-basis difference. Because historical index levels are not retained, the coin-to-USD conversion uses a parity-implied spot as a proxy. The completed textual audit covers all 124 events: rule text is available for every event, but all 60 BTC and 64 ETH events are classified as `proxy_assumed`, with a reference-basis mismatch share of 1.0 for both assets. The audit documents the mismatch but does not change the numerical sample gate. The comparison is therefore systematic and transparent, but it is not an exact settlement-basis match; this limitation is revisited in Chapter 7.

Non-synchronicity at the bar level. Daily and hourly OHLC bars record that an instrument traded within a bar but not the intraday instant of the trade, so a reconstructed cross-strike “snapshot” stitches together trades that need not be simultaneous. This residual non-synchronicity cannot be removed with OHLC data and is quantified only through bar-level staleness; it is the reason the Track B intraday comparison is restricted to the six-hour frequency, at which the estimator noise is materially reduced.

Chapter 5 Results

This chapter reports the empirical findings of the two tracks. Track A (Section 5.1) compares full terminal-price distributions at the daily frequency; Track B (Section 5.2) compares an at-the-money survival probability at intraday frequency. Section 5.3 draws the two together. Throughout, findings are graded by their robustness: results that survive every sensitivity check are stated as such, and results that depend on a modelling choice are reported as contingent on that choice rather than as hard conclusions. The methods underlying each quantity are defined in Chapter 3; the samples and gates are defined in Chapter 4.

5.1 Track A: distribution divergence

The Track A comparison sample contains 294 event-days from 61 events, corresponding to 3114 cell-day rows. For each event-day the Polymarket distribution and the fitted Deribit risk-neutral density are placed on the same event bucket grid, and their difference is summarised both by cell-level distance (L1, L2) and by the location, spread, and tail moments of the two distributions. Figure 5.1 illustrates the common-grid comparison for one event-day. It is included to make the empirical object concrete, not as evidence that this single event represents the cross-event average.

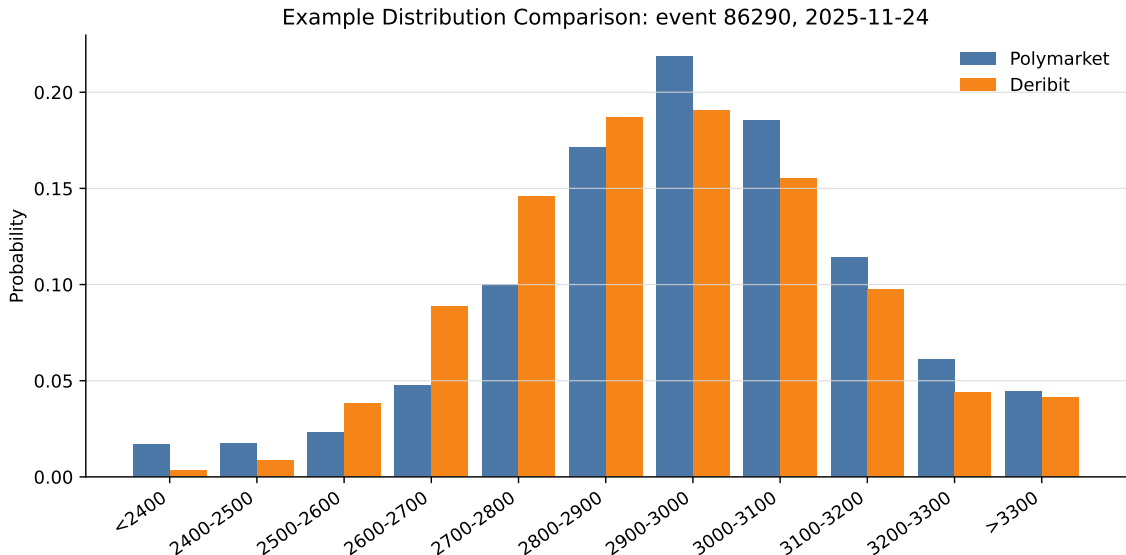


Figure 5.1: Illustrative Track A distribution comparison for event 86290 on 24 November 2025. Both probability vectors are evaluated on the same Polymarket bucket grid. The event-day is an example of the constructed comparison and is not used as a representative-effect estimate.

The final sample is also conditional on the option-curve quality gate. As Figure 5.2 shows, 294 of the 301 successfully fitted event-days pass the coin-denominated repricing-RMSE threshold; the seven failures remain visible in the diagnostic rather than being silently discarded.

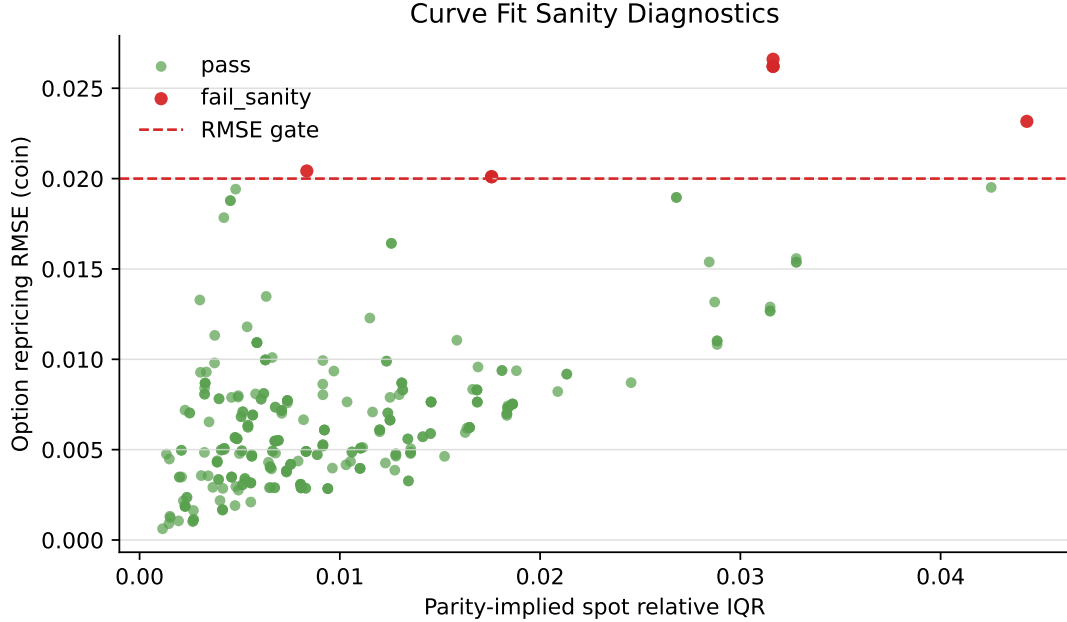


Figure 5.2: Track A curve-fit diagnostic. Points show option-repricing RMSE against the dispersion of strike-level parity-implied spot estimates; the dashed line is the implemented RMSE gate. Colours report the complete curve-quality decision, so a point below the RMSE line can still fail another sanity condition.

5.1.1 The two markets agree on the centre of the distribution

The clearest Track A result is a negative one: the two venues do not disagree about where the terminal-price distribution is centred. Regressing the location difference (Polymarket mean minus Deribit mean) on the signed horizon-gap bins, time-to-expiry, and asset yields an intercept of 0.000 572 that is statistically indistinguishable from zero ($p = 0.8015$), and none of the composition terms is significant; the regression explains almost none of the variation ($R^2 = 0.011$). The absence of both a level shift and a composition pattern means the appropriate framing is not that “Polymarket is systematically more bullish or bearish,” but that the centres align and the disagreement, where it exists, is in the *shape* of the distribution. This is the hard Track A finding.

5.1.2 Polymarket prices a wider distribution under light-to-moderate smoothing

At the baseline risk-neutral density specification, Polymarket terminal distributions are wider than Deribit’s: the median Polymarket spread is 0.0515 against 0.0456 for Deribit, a median difference of 0.0040, and Polymarket is the wider distribution on 70.7% of event-days. Read on its own this is a modest but systematic width wedge in the direction of Polymarket pricing more dispersion.

This ordering is, however, contingent on the smoothness penalty used in the density fit. Table 5.1 reports the width wedge and the in-sample option-repricing error across the penalty grid. The wedge is strongest at the lightest regularisation and attenuates monotonically as the penalty increases, falling to a median difference of 0.0008 and a wider-share of 0.526—close to one half, although no formal test against one half is reported—at the heaviest level examined. Because in-sample repricing error is itself monotone in the penalty (a lighter penalty always fits the traded strikes more closely), it cannot by itself select a regularisation level; a light penalty minimises in-sample error but overfits the sparse strike grid. We therefore do not privilege a single specification and report the width wedge as a *regularisation-contingent* secondary finding: Polymarket is wider under light-to-moderate smoothing, and the wedge attenuates under heavy smoothing.

Table 5.1: Width wedge and in-sample repricing error across the smoothness penalty. Common sample of 293 event-days across 61 events. “PM wider share” is the fraction of event-days on which the Polymarket spread exceeds the Deribit spread; “spread diff” is the median of (Polymarket – Deribit) spread; “RMSE” is the coin-denominated option-repricing error.

Smoothness penalty	RMSE (coin)	PM wider share	Median spread diff
0.00	0.005 925	0.9079	0.009 556
0.05	0.006 166	0.8123	0.006 791
0.10	0.006 351	0.7065	0.003 968
0.20	0.006 669	0.5256	0.000 843

Two checks that do *not* turn on the smoothness penalty confirm that the baseline ordering is not an artefact of the tail treatment or the state grid. Varying the representative value assigned to the open-ended tail buckets from half a bucket width to a full bucket width leaves the ordering essentially unchanged (Polymarket wider on 70.7% versus 71.8% of event-days). The fitted densities place negligible mass at the boundaries of the state grid (median edge mass 0.0014; only 1 of 294 event-days exceeds 5% edge mass), so the width comparison is not driven by truncation of the Deribit density. The sign of the width wedge is thus robust to tail-midpoint and truncation choices among the reported checks; smoothness remains the material qualification to the sign and magnitude.

5.1.3 Disagreement is concentrated in the tails

Relative to the probability mass they carry, the tail cells disagree far more than the interior. At baseline the mean absolute divergence is almost identical in the two regions (0.0269 in the tails, 0.0273 in the body), but because tail cells carry roughly half the probability mass of body cells, the same absolute divergence corresponds to a much larger relative disagreement: mean relative absolute divergence of 0.858 in the tails against 0.568 in the body, and a median log-odds divergence of 1.42 in the tails against 0.004 in the body. The direction is that Polymarket assigns more mass to the extreme buckets, consistent with a favourite-longshot pattern. Because raw tail probabilities are mechanically small, the tail wedge is reported using relative and log-odds metrics rather than absolute probability differences. The relative tail wedge remains larger than the body wedge across the smoothed penalty grid: tail relative absolute divergence is 0.889, 0.857, and 0.944 for smoothness penalties 0.05, 0.10, and 0.20, respectively, against body values of 0.514, 0.571, and 0.754. The sign of the tail result is therefore more stable than the width wedge, while the exact magnitude remains sensitivity-dependent.

5.1.4 Shape distance is present but secondary

The cell-level distribution distance is non-trivial—median normalised L1 of 0.235 and median L2 of 0.090—but it is the quantity most sensitive to the smoothness penalty: removing the penalty inflates the median L1 to 0.885. This jump is not interpreted as a substantive fourfold increase in economic disagreement. Rather, an unsmoothed fit to sparse strikes produces a jagged density whose cell-by-cell distance from the smooth Polymarket distribution is large even when lower-dimensional summaries, especially the location moment, remain similar. L1 and L2 are therefore reported as secondary shape-distance diagnostics rather than as headline magnitudes.

5.1.5 The horizon-gap gradient is a composition control, not a maturity effect

The width wedge varies strongly and monotonically with the signed horizon gap between the Polymarket resolution and the matched Deribit expiry. Relative to the -8 hour reference bin, the spread-difference regression gives coefficients of -0.0041 ($-32h$), 0.0067 ($+16h$), and 0.0094 ($+40h$), all significant with event-clustered standard errors, together with an ETH effect of 0.0054 and a time-to-expiry effect of 0.0011 ($R^2 = 0.468$). The gradient survives partialling out asset and time-to-expiry (partial Spearman of 0.653) and holds within each asset. Figure 5.3 displays the corresponding unconditional event-day distributions. The visual gradient is descriptive: the bins are calendar-alignment categories, not assigned maturity treatments.

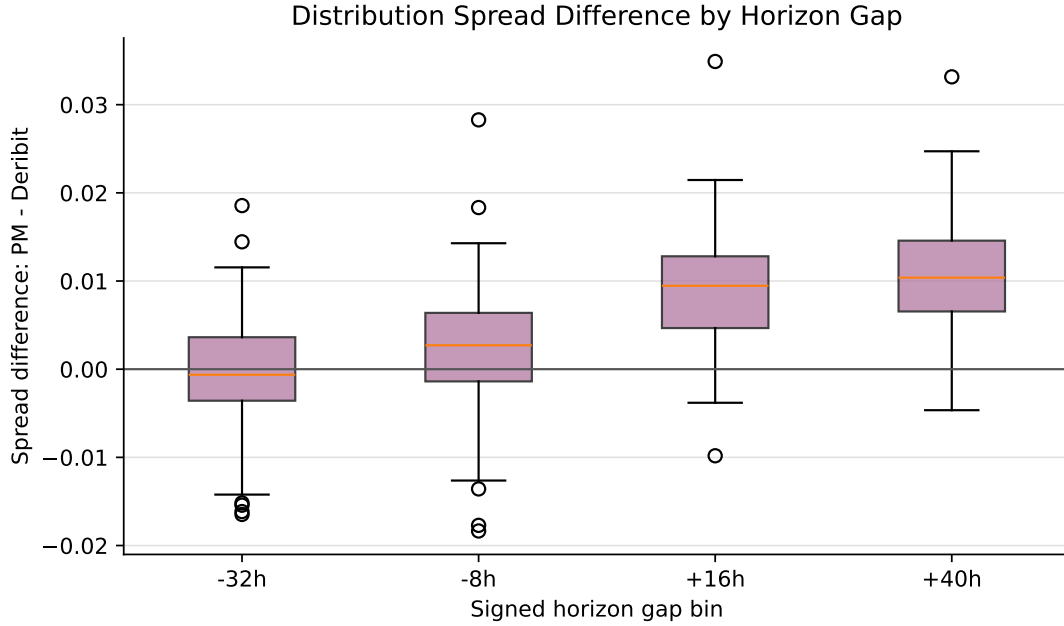


Figure 5.3: Track A spread difference by signed horizon-gap bin. Positive values indicate a wider Polymarket distribution. The groups describe calendar and expiry composition and must not be interpreted as causal maturity effects.

This gradient must not be interpreted as a causal maturity effect. Its sign is inconsistent with a simple diffusion-horizon story: conditioning on the Deribit time-to-expiry, a longer Polymarket horizon does not widen the Polymarket distribution in the way diffusion would predict. The signed gap is instead a deterministic function of the calendar offset between the two venues' settlement dates, so it is best read as a calendar/expiry-alignment composition control. The precise mechanism—whether it reflects the type of the nearest Deribit expiry or the composition of Polymarket market types—is not identified by the available data and is left to Chapter 7.

5.2 Track B: cross-market integration and price discovery

Track B compares the survival probability $P(S_T > K^*)$ at a single near-the-money reference strike. The primary sample is the 79 bucket-distribution events; point-threshold events are excluded because their survival series are saturated near zero or one on 42.1% of hours (against 9.0% for bucket-distribution events) and therefore carry too little time-series variation for a lead-lag comparison, whatever their superficial quality.

5.2.1 Hourly changes are not reliable for directional inference; levels agree

At the hourly frequency the two venues agree on the *level* of the survival probability but not on its *changes*. Across joint-informative hours the Deribit local-survival change has roughly three times the standard deviation of the Polymarket change and a strongly negative first-order autocorrelation, the signature of microstructure/estimation noise in a call-spread digital built from two sparse traded strikes. The contemporaneous correlation of hourly changes is therefore near zero, and after joining the two venues only 2973 joint-informative hours remain. Only 2 events retain at least 72 total joint-informative hours, and no event has a contiguous joint-informative run of 24 hours or more. The hourly panel is consequently a feasibility diagnostic, not a frequency at which directional price discovery can be measured with the current local-digital estimator and observed strike coverage.

5.2.2 At six hours the two markets co-move

Re-estimating the Deribit digital on six-hour bars averages down the estimation noise and restores a measurable common signal. At six hours the survival *levels* are highly aligned (correlation 0.912, median absolute wedge 0.049), and the *changes* co-move contemporaneously with a correlation of 0.535 across 889 contemporaneous change pairs from 78 events. The Deribit-to-Polymarket change standard-deviation ratio falls to 1.59 and the negative autocorrelation weakens. Both changes are consistent with lower measurement noise at the coarser frequency, although aggregation also increases within-block non-synchronicity. The two markets are, on this evidence, integrated around the reference strike: their at-the-money survival probabilities are strongly aligned but not identical, and they move together at six-hour frequency.

5.2.3 Directional leadership is not identified

The natural next question—which venue leads—cannot be answered with these data, and the apparent evidence that it can is an artefact. The symmetric cross-correlation shows no robust lead in either direction: the six-hour lead correlations are 0.074 (Deribit leading) and 0.025 (Polymarket leading), both dwarfed by the contemporaneous 0.535; the frozen analysis contains no inference establishing either lead correlation as different from zero. The pooled lag regressions appear to tell a different story—the Polymarket-lag coefficient in the Deribit equation is 0.330 ($p < 0.001$) while the Deribit-lag coefficient in the Polymarket equation is only 0.058 ($p = 0.062$)—but this asymmetry is what differential measurement error produces, not evidence of Polymarket leadership. When the noisy Deribit change is the right-hand-side regressor, its coefficient is attenuated by

errors-in-variables; when the cleaner Polymarket change is the regressor and the noisy Deribit change is the dependent variable, the coefficient is not attenuated in the same way. The Deribit own-lag coefficient of -0.401 is consistent with substantial short-horizon reversal or estimation noise remaining even at six hours. These p-values use asymptotic event-clustered standard errors; a wild-cluster bootstrap is not part of the frozen analysis. More importantly, the regression and symmetric diagnostics do not provide concordant directional evidence. We therefore report directional leadership as *unidentified*. The robust Track B result is contemporaneous integration at six-hour frequency, with sub-six-hour price discovery unidentifiable given the liquidity of the available Deribit option data.

5.3 Synthesis

The two tracks describe the same relationship from different angles. Track A shows that Polymarket and Deribit do not disagree about the centre of the BTC/ETH terminal-price distribution; the divergence is in the shape—Polymarket prices a wider, more tail-heavy distribution under light-to-moderate smoothing. Track B shows that this shape disagreement does not reflect two disconnected markets: around a near-the-money threshold the two venues’ survival probabilities are highly aligned in level and co-move contemporaneously at six-hour frequency. The combined picture is one of *integration with a shape wedge*: the markets track the same events closely, and the residual disagreement is best framed as a probability-measure and tail-risk wedge under liquidity constraints, rather than as a mechanical arbitrage or a one-sided information advantage. The remaining questions—which venue leads at sub-six-hour horizons, and what precisely drives the calendar-alignment gradient—are limited by the granularity and liquidity of the option data rather than by the comparison design, and are taken up in Chapters 6 and 7.

Chapter 6 Discussion

The results support a more qualified interpretation than either market equivalence or market disconnection. Polymarket and Deribit agree closely on the centre of the BTC/ETH terminal-price distribution and on the local level of a near-the-money survival probability. At the same time, they differ more materially in relative tail pricing, and Polymarket appears wider under low-to-moderate regularisation of the Deribit state-price fit. Intraday changes co-move at six hours, but the available data do not identify a leading venue. Taken together, these findings describe frequency-bounded integration with a shape wedge.

This chapter explains that interpretation in relation to the research questions and prior literature. It first considers why centre agreement can coexist with tail disagreement, then discusses the regularisation-dependent spread result and the horizon-gap gradient. It next interprets the Track B integration evidence and unidentified direction of price discovery. The final sections consider alternative explanations, economic significance, and implications for future research design.

6.1 Centre agreement and shape disagreement

The near-zero location result is economically informative because it rules out the simplest account of the cross-market wedge. If Polymarket traders were persistently more optimistic or pessimistic about BTC and ETH than option-market participants, the Polymarket distribution would be shifted relative to Deribit's. The controlled location intercept of 0.000 572, with $p = 0.8015$, provides no evidence of such a systematic shift. Nor do the observed horizon-gap, asset, and time-to-expiry controls explain meaningful location variation. The two markets therefore appear to share a common assessment of the central terminal-price region.

Agreement at the centre does not imply equality of distributions. Two probability vectors can have similar means while assigning different mass to the shoulders and tails. That is the pattern observed here. The average tail-relative absolute divergence is about 0.86, compared with roughly 0.57 in the body. The corresponding absolute cell differences are much closer because tail probabilities are small by construction. Relative and log-odds comparisons are consequently more informative than raw percentage-point differences for judging tail disagreement.

One interpretation is a favourite-longshot mechanism. Prediction-market research shows that low-probability outcomes can be overpriced relative to their realised frequency or benchmark probability, with probability misperception offering one explanation

(Snowberg and Wolfers, 2010). Portnaya’s pointwise crypto evidence likewise finds the largest Polymarket-option wedge at low option-implied probabilities (Portnaya, 2026). The present full-distribution result is consistent with those findings: the disagreement is not a uniform directional shift but is concentrated in low-mass regions.

Consistency is not identification, however. The Deribit vector is risk-neutral, whereas the Polymarket vector is an event-market price vector shaped by beliefs, risk preferences, collateral, and liquidity. Option-implied probabilities embed a pricing kernel and need not equal physical probabilities (Jackwerth, 2004). A larger Polymarket tail probability could therefore reflect longshot demand, different subjective beliefs, an option-market variance risk premium, or different compensation for liquidity and settlement risk. The current design identifies where the distributions differ, but not which of these mechanisms creates the difference or which probability is closer to the physical data-generating process.

6.2 Why the spread result remains conditional

At the baseline state-price specification, the Polymarket distribution is wider on 70.7% of event-days and the median spread difference is 0.0040. The direction is stable when the open-tail midpoint convention is changed and when state-grid edge mass is inspected. It is not stable to the smoothness row weight. On the common sample, the median difference declines from 0.0096 with no smoothness penalty to 0.0008 at the heaviest setting, where the wider share is 0.526. Repricing RMSE increases only modestly across this grid, so in-sample fit alone cannot reject the heavy specification.

This sensitivity follows from the inverse problem rather than from an incidental coding choice. The number of discrete state intervals is similar to the number of usable option strikes. Without regularisation, a non-negative least-squares fit can reproduce local strike noise with a jagged state vector. Increasing the penalty smooths those local oscillations and shifts probability mass across neighbouring Polymarket cells. Location is comparatively stable because local mass reallocations have limited effect on the first moment; spread and L1 distance are more exposed because they respond directly to how mass is distributed away from the centre.

The defensible conclusion is therefore not that Polymarket is unconditionally more dispersed. It is that a width wedge is present under the baseline and lighter regularisation choices but attenuates under heavy smoothing. This qualification also explains why L1 and L2 are secondary diagnostics. Their magnitudes partly measure disagreement between venues and partly measure the roughness permitted in the recovered Deribit distribution. A different low-dimensional estimator, such as a parametric mixture, would be needed before promoting a precise cell-level distance to a model-invariant economic effect.

The horizon-gap gradient is statistically clearer but not causally cleaner. The spread

difference rises across the signed calendar-alignment bins after controlling for asset and Deribit time to expiry. Yet the gap is determined by the alignment of Polymarket resolution dates with listed Deribit expiries. It is not assigned independently and can proxy for expiry type, weekday, event composition, or other market conditions. Its sign is also inconsistent with a simple account in which a mechanically longer horizon alone creates the wider distribution. The gradient is useful as evidence that matching structure matters; it is not an estimate of the effect of adding an hour to one contract’s maturity.

6.3 Frequency-bounded integration

Track B changes the object from a complete daily distribution to one local survival probability. At the hourly frequency, the level series are aligned, but Deribit changes are much noisier and strongly negatively autocorrelated. Sparse fresh call strikes, same-bar parity requirements, and last-trade OHLC create a local digital that is too unstable for directional inference. This is a measurement limitation rather than evidence that the underlying markets fail to communicate.

Six-hour construction improves the availability of bracketing calls and parity pairs. In the resulting informative sample, the survival-level correlation is 0.912, and the contemporaneous change correlation is 0.535. These values support RQ2: the markets are integrated around the selected threshold at the observable six-hour frequency. Integration here means that the two local probabilities share a strong common component and adjust within the same six-hour block. It does not mean equality: the median absolute wedge remains about five percentage points, and each block can combine instrument closes from different hours.

The distinction matters for price discovery. The symmetric cross-correlation is largest contemporaneously. The Deribit-leading and Polymarket-leading six-hour correlations are only 0.074 and 0.025, respectively, and the frozen analysis contains no inference establishing either as different from zero. Pooled event-fixed-effect regressions appear more directional: lagged Polymarket changes predict Deribit changes, while the reverse coefficient is small and marginal. But Deribit changes remain about 1.59 times as volatile and have a lag-one autocorrelation of -0.248 . Classical errors-in-variables logic then attenuates the coefficient when noisy Deribit changes appear on the right-hand side, while no equivalent attenuation applies when they are the dependent variable.

This is the noise-contamination problem emphasised by the price-discovery literature (Putniņš, 2013; Patel et al., 2020). A venue can appear to lead because the other price measure is noisier, not because information reaches it first. The regression p-values also rely on asymptotic event-clustered standard errors; a wild-cluster bootstrap is not part of the frozen analysis. The symmetric and regression diagnostics therefore do not provide concordant evidence for a ranking. RQ3 is answered as unidentified, and Hasbrouck or

information-leadership shares would add structure unsupported by the available continuous runs and measurement quality.

This result does not contradict Ng et al. (2026), who find that Polymarket leads Kalshi when liquidity and activity are high. Their comparison involves common election contracts across prediction-market platforms. Here one side is a constructed option-implied digital based on sparse traded strikes. The difference in securities and measurement makes leadership harder to observe and is itself evidence that price-discovery methods cannot be transferred mechanically across settings.

6.4 Alternative explanations and economic meaning

Several mechanisms can produce the observed shape wedge without inconsistent information. First, all 124 canonical events fail an exact textual reference-basis match: the completed audit classifies every event as `proxy_assumed`. Polymarket resolves against a centralised-exchange price specified in its rules, while Deribit settles against its own index. Even when the underlying asset and date match, basis risk remains near the terminal boundary.

Second, resolution and option-expiry times differ by 8–40 hours. This matters most in the tails, where a short additional horizon can materially change threshold probabilities. The regression controls observed alignment, but it cannot convert the two securities into identical claims. Third, the parity-implied spot abstracts from rates, funding, and historical index values. Those approximations are small candidates over short horizons but are not measured away.

Fourth, the final Track A sample is selected toward more liquid events. The six Polymarket-quality events removed entirely before curve fitting have substantially lower median event volume than the 61 survivors. It is plausible that segmentation is larger in excluded events, but it is also plausible that their apparent wedges would be dominated by stale-price error. The direction of selection bias is therefore unknown.

These points prevent a law-of-one-price or arbitrage interpretation. The Polymarket claim and a portfolio of Deribit options are not fungible: they can settle at different times and references, require different collateral, and expose traders to different liquidity and execution risks. Research on Polymarket documents both limited depth for executable arbitrage (Cheng et al., 2026) and off-chain matches that later revert on chain (Shen et al., 2026). Cross-platform work also finds persistent execution-aware deviations among semantically similar but non-identical markets (Gebele and Matthes, 2026). The statistical wedge in this dissertation is therefore a pricing and integration result, not a backtested profit opportunity.

6.5 Implications

For empirical research, the main implication is that data improvement has higher value than estimator complexity at this stage. Trade- or quote-level Deribit data would permit genuinely simultaneous strike snapshots, direct index matching, and sub-six-hour adjustment tests. Exact archived settlement prints would turn the textual reference audit into a numerical basis analysis. An alternative low-dimensional risk-neutral density would reveal whether the tail result and conditional width result survive a different regularisation architecture.

For market interpretation, the results show that integration can be object-specific. Polymarket and Deribit can share a central probability signal and move together around a near-the-money threshold while still pricing relative tail mass differently. Treating market integration as a binary property would miss this distinction. The appropriate conclusion is narrower: the venues are strongly integrated at the centre and at six-hour frequency, but distributional identity and directional leadership are not established.

Chapter 7 Limitations and Conclusion

This dissertation compares prediction-market event probabilities with option-implied risk-neutral probabilities for matched BTC/ETH terminal-price events. Chapter 6 interprets the evidence as frequency-bounded integration with a shape wedge. This chapter separates the limitations of that interpretation from the final conclusion.

Those conclusions are intentionally narrower than the original project prompt. The evidence supports *cross-market probability divergence with frequency-bounded integration*. It does not support a mechanical arbitrage claim, a venue-leadership ranking, or a single structural explanation for the observed wedge. This chapter states the main limitations before drawing the final conclusion.

7.1 Measurement limitations

7.1.1 Polymarket and Deribit probabilities are not the same object

The central comparison is economically meaningful but not frictionless. Polymarket prices are event-market probabilities under a physical, collateral, liquidity, and trader-belief environment. Deribit option prices imply risk-neutral state prices under a derivatives-market environment. A difference between them therefore mixes beliefs, risk premia, margin and funding costs, segmentation, and model error. Throughout the dissertation the word “divergence” is used for this reason: a wedge is not automatically a mispricing, and it is not automatically tradable.

This distinction is especially important for the Track A tail results. A larger relative tail wedge is consistent with a favourite–longshot style pattern in prediction-market probabilities, but the dissertation cannot decompose that pattern into risk-loving preferences, subjective tail beliefs, hedging demand, or liquidity premia. The empirical claim is that the two venues price the tails differently on the constructed common grid, not that one venue is objectively wrong.

7.1.2 Settlement reference and horizon mismatch

The event matching is necessarily approximate. The Polymarket resolution instant and the nearest Deribit expiry differ by 8–40 hours. Chapter 5 shows that the signed horizon-gap bins explain a large share of the spread variation, but their sign is inconsistent with a simple causal maturity interpretation. They are therefore treated as calendar/expiry-

alignment controls rather than as identified maturity shocks.

The reference basis is also imperfect. Polymarket event rules resolve against the reference specified in the market text, while Deribit settles against its own index. The completed textual audit covers all 124 canonical events and finds rule text available for every event. It classifies all 60 BTC and 64 ETH events as `proxy_assumed`: both references are documented, but the reference-basis mismatch share is 1.0 for each asset. Historical Deribit index levels are not retained in the raw OHLC files, so the pipeline uses a parity-implied spot proxy for the coin-to-USD conversion. The audit therefore turns an unknown mapping into a documented mismatch; it does not establish exact reference-basis identity or remove numerical basis risk.

7.1.3 OHLC bars are not simultaneous snapshots

Both Track A and Track B rely on historical OHLC rather than full trade and quote histories. A daily Deribit option surface stitches together last traded prices from different strikes within the same day. An hourly or six-hour local digital does the same within an intraday bar. The bar time stamp is therefore not a true cross-strike snapshot. Daily Track A controls this with staleness filters, while Track B avoids a full intraday density and estimates only a local call-spread digital around K^* .

This limitation matters directly for price discovery. At the hourly frequency, the Deribit local-survival changes are too noisy: levels are aligned, but changes are dominated by microstructure and sparse-strike measurement error. Aggregating to six-hour bars restores a measurable common signal, but it also rules out identification of sub-six-hour leadership. The resulting Track B conclusion is therefore integration, not leadership.

7.2 Model and estimator limitations

7.2.1 Risk-neutral density recovery is regularisation-dependent

The Track A estimator follows the Breeden–Litzenberger logic only after smoothing and no-arbitrage discipline. It does not apply raw second differences to discrete option prices. This is a necessary choice: Deribit strike grids are sparse and stale enough that raw finite differences would convert last-trade noise into negative or spiky densities.

The cost is model dependence. The smoothness penalty is load-bearing because the state grid and the option strike grid are of comparable size. The most important consequence is the Track A spread result. Under low-to-moderate smoothing, Polymarket distributions are wider than Deribit-implied distributions. Under the heaviest smoothing in the grid, the median spread difference falls close to zero and the Polymarket-wider share becomes close to one half; no formal test against one half is reported. The option

repricing error does not worsen enough at that heavy setting to exclude it on fit-quality grounds alone. For this reason the dissertation reports the spread wedge as smoothing-conditional, not as a hard sign-invariant result.

The centre and tail-relative results are safer. The location difference remains close to zero across the smoothness grid, and the relative tail wedge remains larger than the body wedge under the smoothed specifications. Even there, the exact magnitude should be read as estimator-dependent. A parametric robustness check, such as a low-dimensional lognormal-mixture density, would be useful if a future version of the dissertation wanted to make cell-level shape distance or a hard spread magnitude the headline result.

7.2.2 Generated regressors and small event clusters

Several Track A variables are generated from fitted Deribit densities rather than observed directly. The spread, tail, and L1 measures inherit uncertainty from the option-surface fit. The reported event-clustered standard errors handle within-event dependence in the second-stage regressions, but they do not fully propagate first-stage RND uncertainty. The regressions should therefore be read as diagnostic controls for observed composition, not as structural estimates.

The number of independent events is also modest: Track A’s main comparison uses 61 events, and Track B’s primary bucket sample uses 79. This is enough to support descriptive event-level inference, but it is not large enough to justify fragile sub-sample claims. In particular, the dissertation rejects making the -8 hour horizon-gap slice the headline sample because that would trade a seemingly cleaner timing match for weaker event-level inference and a less representative comparison.

7.3 Identification limitations

7.3.1 The horizon-gap gradient is not causally identified

The signed horizon gap is one of the strongest Track A correlates of the spread wedge. However, it is not randomly assigned and should not be interpreted as a causal maturity effect. The gap is determined by how Polymarket resolution dates line up with the nearest Deribit expiry. It may proxy for expiry type, calendar-day alignment, or other event-composition features. The regression therefore uses it as a control for observed matching structure, not as evidence that increasing one venue’s horizon mechanically changes the wedge.

7.3.2 Directional price discovery is unidentified

Track B was originally designed to ask which venue reacts first. The final data support a narrower conclusion. At six hours, Polymarket and Deribit local survival probabilities are highly aligned in levels and co-move strongly in changes. But the symmetric lag correlations show no robust lead in either direction, and the pooled lag regressions are contaminated by differential measurement error. The larger Polymarket-lag coefficient in the Deribit equation cannot be read as Polymarket leadership because Deribit changes are the noisier series. When the noisy series appears on the right-hand side, the coefficient is attenuated; when it appears on the left-hand side, the coefficient is not attenuated in the same way.

The dissertation therefore reports directional leadership as unidentified. This is a substantive limitation, but it is also a useful empirical result: with public OHLC data, the price-discovery object is frequency-bounded. Hourly data are too noisy, six-hour data are integrated but too coarse for sub-six-hour leadership, and the available panel does not support a Hasbrouck or per-event VECM decomposition.

7.4 External validity and economic interpretation

7.4.1 The sample is narrow by design

The canonical universe consists of 124 BTC/ETH terminal price events. These events are valuable because they are templated, repeated, and matchable to listed crypto options. They are not representative of all Polymarket markets, all prediction-market topics, or all crypto derivatives. The results should therefore be interpreted as evidence for liquid BTC/ETH terminal price events, not as a general statement about prediction markets.

The event horizon is also short, with most events living around one week. The findings may not transfer to long-horizon political contracts, sports markets, or low-liquidity thematic markets. Those settings have different trader bases, settlement rules, collateral costs, and liquidity conditions.

7.4.2 No trading-profit claim is made

The dissertation does not estimate executable trading profits. This is deliberate. A statistical probability wedge is not sufficient for an arbitrage strategy because the claims are non-fungible, settlement references may differ, both venues have fees and collateral constraints, and Deribit options hedge a continuous payoff while Polymarket contracts settle as binary event claims. The results are therefore framed as market-integration and probability-divergence evidence, not as a Sharpe ratio or relative-value backtest.

An economically implementable extension would need executable order-book depth, fees, funding, collateral haircuts, slippage, and settlement-reference risk. It would also need a decision rule that is fixed before looking at realised outcomes. Those requirements are outside the P1 empirical scope.

7.5 Future work

The most valuable extensions are data improvements rather than more complicated estimators on the same OHLC panel.

- **Numerical settlement-basis reconstruction.** The textual audit is complete and shows that all events are proxy rather than exact matches. The next step is to archive the actual Polymarket settlement print and contemporaneous Deribit index level for each event, then quantify how much of the observed wedge can be attributed to the basis difference near expiry.
- **Trade- or quote-level Deribit data.** A true intraday option snapshot would separate stale last-trade noise from price discovery. With denser data, the Track B question could return to sub-six-hour leadership and, if the time-series assumptions are satisfied, to information-share or information-leadership methods.
- **Parametric RND robustness.** A low-dimensional risk-neutral density, for example a lognormal mixture, could test whether the tail and spread results survive a different form of regularisation. This is most useful if future work wants to make cell-level distribution distance a primary result.
- **Broader event families.** Extending beyond BTC/ETH terminal price templates would test whether the observed probability wedge is a crypto-price phenomenon or a broader prediction-market feature.
- **Execution-aware economic value.** A separate trading chapter should only be added with executable quotes, depth, fees, and fixed ex-ante rules. It should test whether the measured wedge is economically tradable, rather than assuming that statistical divergence is an arbitrage.

Chapter 8 Conclusion

The dissertation’s main result is not that one market is correct and the other is wrong. Nor is it that one venue consistently leads the other. The evidence is more precise: Polymarket and Deribit are integrated around the same BTC/ETH terminal price events, but they are not distributionally identical.

Track A shows that the centres of the terminal distributions align. The robust disagreement is instead in distributional shape. Tail-relative divergence is material, and the Polymarket width wedge is visible under low-to-moderate RND smoothing but attenuates under heavy smoothing. Track B shows that this distributional disagreement does not arise from two disconnected markets. Around a near-the-money threshold, the two venues’ survival probabilities are highly aligned in levels and co-move at the six-hour frequency. What cannot be recovered from the current OHLC data is a clean directional ranking of price discovery.

The final interpretation is therefore *frequency-bounded integration with a shape wedge*. The markets track the same underlying terminal price events, but they price tail risk and distributional width differently under the constraints of different probability measures, settlement references, liquidity, and market microstructure. That is the defensible contribution of the empirical design: it turns an initially broad “order-book divergence” question into a reproducible cross-market probability comparison with explicit limits on what the data can and cannot identify.

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Chapter A Appendix

Put your appendix materials here.

Project Summary

Put here at the end of the thesis a copy of the industrial project summary.